

Module - 03

Network Scanning

→ Network Scanning: It refers to set of procedures used for identifying hosts, ports, and services in a network.

→ TCP Communication Flags:

- SYN – Synchronize Packet, Client sends a request to server.
- SYN/ACK – Server Acknowledge client's request.
- ACK – Client Acknowledge Server's request.

[Three-way Handshake]

- Urg – Urgent, Packet should be processed immediately.
- FIN – No further transmission.
- RST – Reset the connection
- PSH – Sends all buffered data immediately.

→ Scanning Tools: Scanning Tools are used to identify live hosts, open ports, running services on a target network, location info, NetBIOS info, and Information about all TCP/IP and UDP open Ports.

→ Scanning Tools are:

- Nmap
- Advanced IP Scanner Tool
- Netdiscover
- NetScanTools Pro

✚ It is the process of discovering live hosts, open ports and services in a network to identify potential entry points.

Host Discovery - Checking for Live Systems

Host Discovery techniques are used to identify the active/live systems in the network.

1. Netdiscover – an active and passive ARP reconnaissance tool used to discover live hosts on a local network.

Command: netdiscover -r 192.168.1.1/24

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.29.69	d2:72:c5:29:fe:1a	7	420	Unknown vendor
192.168.29.1	f0:ed:b8:48:01:78	287	17220	SERVERCOM (INDIA) PRIVATE LIMITED
192.168.110.130	00:0c:29:3d:78:90	218	13080	VMware, Inc.
192.168.29.85	50:c2:e8:90:89:ad	4367	262020	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.29.10	4a:d2:b3:c3:25:7b	1	60	Unknown vendor
192.168.29.109	00:0c:29:3d:78:90	20	1200	VMware, Inc.
192.168.110.2	00:50:56:fd:56:9b	85	5100	VMware, Inc.
192.168.110.254	00:50:56:f4:8b:36	15	900	VMware, Inc.
192.168.110.1	00:50:56:c0:00:08	372	22320	VMware, Inc.

2. Advance IP Scanner Tool: Target = 192.168.110.130

Status	Name	IP	Manufacturer	MAC address	Comments
	reliance.reliance	192.168.29.1		F0:ED:B8:48:01:78	
		192.168.29.8	VMware, Inc.	00:0C:29:72:F4:4C	
		192.168.29.10		4A:D2:B3:C3:25:7B	
		192.168.29.39		A6:FF:52:F5:30:8D	
		192.168.29.69		D2:72:C5:29:FE:1A	
	Cipher-Dusk	192.168.29.85	CLOUD NET...	50:C2:E8:90:89:AD	
		192.168.29.136		EE:9A:4C:7E:4E:00	
		192.168.29.223		94:A9:90:65:DF:F4	
	Cipher-Dusk	192.168.110.1	VMware, Inc.	00:50:56:C0:00:08	
		192.168.110.130	VMware, Inc.	00:0C:29:3D:78:90	
		192.168.110.254	VMware, Inc.	00:50:56:F4:8B:36	
	Cipher-Dusk	192.168.232.1	VMware, Inc.	00:50:56:C0:00:01	
		192.168.232.254	VMware, Inc.	00:50:56:EB:53:58	

ICMP, ARP - Checking for Live Systems

ICMP: Internet Message Control Protocol, test connectivity and reports network errors.

Ping = Packet InterNet Groper [Network Tool]

ICMP Ping Scan Process:

- Client Sends ICMP Echo request to host.
- If host is alive, it will return ICMP Echo reply.

ARP: Address Resolution Protocol, resolves IP into MAC, works only on LAN, used by Nmap for Local Host Discovery

- Command: `nmap -sn 192.168.1.1/24`
-sn = Host Discovery Only

```
(root@kali)-[~]  
# nmap -sn 192.168.110.130  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-01 07:04 -0400  
Nmap scan report for 192.168.110.130  
Host is up (0.0011s latency).  
MAC Address: 00:0C:29:3D:78:90 (VMware)  
Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds
```

Nmap

Nmap: Nmap is used to extract information such as Live hosts on network, services (application, name and version), type of packet filters/firewalls, operating systems and OS version.

- Target Machine: Metasploitable2

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:3d:78:90
          inet addr:192.168.110.130  Bcast:192.168.110.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe3d:7890/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:90 errors:0 dropped:0 overruns:0 frame:0
          TX packets:88 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:7334 (7.1 KB)  TX bytes:10653 (10.4 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:154 errors:0 dropped:0 overruns:0 frame:0
          TX packets:154 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:50237 (49.0 KB)  TX bytes:50237 (49.0 KB)
```

- Command: nmap 192.168.110.130

```
(root@kali)-[~]
# nmap 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 08:33 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.94 seconds
```

▪ Command: `nmap -sV 192.168.110.130`

[Version]

```
(root@kali)~  
# nmap -sV 192.168.110.130  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 08:40 -0400  
Nmap scan report for 192.168.110.130  
Host is up (0.0031s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1  
MAC Address: 00:0C:29:3D:78:90 (VMware)  
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.53 seconds
```


■ Command: nmap -sC 192.168.110.130

[Script]

```

root@kali: ~
Session Actions Edit View Help

root@kali: ~ x root@kali: ~ x root@kali: ~ x

nmap -sC 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 08:49 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0038s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.110.128
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh
|_ ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet
25/tcp    open  smtp
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ sslv2:
|   SSLv2 supported
|_ ciphers:
|   SSL2_RC2_128_CBC_WITH_MD5
|   SSL2_DES_64_CBC_WITH_MD5
|   SSL2_RC4_128_EXPORT40_WITH_MD5
|   SSL2_DES_192_EDE3_CBC_WITH_MD5
|   SSL2_RC4_128_WITH_MD5
|   SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_ smtp-command: Metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ ssl-date: 2026-06-30T12:51:10+00:00; +1s from scanner time.
53/tcp    open  domain
|_ dns-nsid:
|_ bind.version: 9.4.2
80/tcp    open  http
|_ http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind
|_ rpcinfo:
|   program version port/proto service
|   100000 2 111/tcp rpcbind
|   100000 2 111/udp rpcbind
|   100003 2,3,4 2049/tcp nfs
|   100003 2,3,4 2049/udp nfs
|   100005 1,2,3 37515/udp mountd
|   100005 1,2,3 38687/tcp mountd
|   100021 1,3,4 41560/tcp nlockmgr
|   100021 1,3,4 44284/udp nlockmgr
|   100024 1 49963/tcp status
|   100024 1 51484/udp status
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec

```

```

root@kali: ~
Session Actions Edit View Help

root@kali: ~ x root@kali: ~ x root@kali: ~ x

1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
|_ mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: Speaks41ProtocolNew, ConnectWithDatabase, Support41Auth, SupportsCompression, SupportsTransactions, SwitchToSSLAfterHandshake, LongColumnFlag
|   Status: Autocommit
|   Salt: R63-VL-Ly/ph9KK-<'M.
5432/tcp  open  postgresql
|_ ssl-date: 2026-06-30T12:50:14+00:00; 0s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
5900/tcp  open  vnc
|_ vnc-info:
|   Protocol version: 3.3
|   Security types:
|   VNC Authentication (2)
6000/tcp  open  X11
6667/tcp  open  irc
|_ irc-info:
|   users: 1
|   servers: 1
|   lusers: 1
|   lservers: 0
|   server: irc.Metasploitable.LAN
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 0:27:43
|   source ident: nmap
|   error: Closing Link: kcwefvpa[192.168.110.128] (Quit: kcwefvpa)
8009/tcp  open  ajp13
|_ ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp  open  unknown
|_ http-title: Apache Tomcat/5.5
|_ http-favicon: Apache Tomcat
MAC Address: 00:0C:29:3D:78:90 (VMware)

Host script results:
|_ clock-skew: mean: 1h00m00s, deviation: 1h59m59s, median: 0s
|_ smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_ System time: 2026-06-30T08:49:57-04:00
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time: Protocol negotiation failed (SMB2)
|_ nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

Nmap done: 1 IP address (1 host up) scanned in 74.91 seconds

```

- Command: `nmap -sC -p 21 192.168.110.130`

[FTP]

```
(root@kali)-[~]
# nmap -sC -p 21 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 10:40 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0013s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 192.168.110.128
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.97 seconds
```

- Command: `nmap -sC -p 22 192.168.110.130`

[SSH]

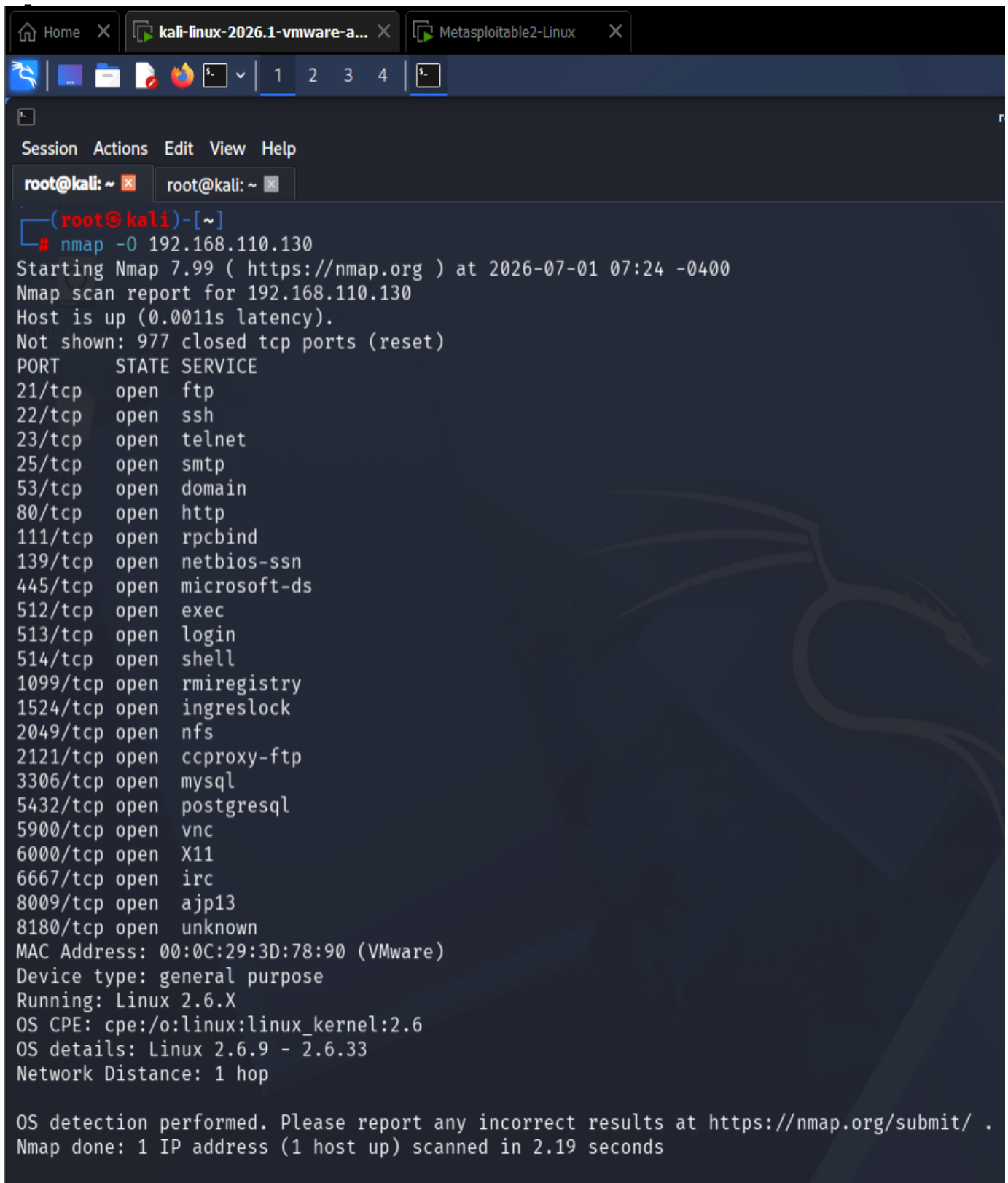
```
(root@kali)-[~]
# nmap -sC -p 22 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-30 11:31 -0400
Nmap scan report for 192.168.110.130
Host is up (0.00077s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.93 seconds
```

- Command: `nmap -O 192.168.110.130`

[Operating System]



```
(root@kali)-[~]
# nmap -O 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-01 07:24 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:3D:78:90 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.19 seconds
```


Firewall / IDS and Evasion using Nmap

- Firewall: Firewall is a security device or software that filters incoming and outgoing traffic to block unauthorized access.
- IDS: Intrusion Detection System, it detects suspicious traffic, generate alerts and monitor network behavior.
- Evasion Techniques: Nmap provides options that make scans harder to detect or classify.

1. MAC Address Spoofing (--spoof-mac)

Changes the source MAC address on local network.

- Command: `nmap --spoof-mac 00:11:22:33:44:55 192.168.110.130`

```
(root@kali)-[~]
# nmap --spoof-mac 00:11:22:33:44:55 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-01 07:56 -0400
Spoofing MAC address 00:11:22:33:44:55 (Cimsys)
Nmap scan report for 192.168.110.130
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.85 seconds
```

2. Decoy Scan (-D)

Adds fake source IP addresses to make the true scanner less obvious.

- Command: `nmap -F -D RND:5 192.168.110.130`
RND = Random IP Addresses
-F = Fast Scan

```
(root@kali)-[~]
# nmap -F -D RND:5 192.168.110.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-01 08:02 -0400
Nmap scan report for 192.168.110.130
Host is up (0.0030s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 00:0C:29:3D:78:90 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.26 seconds
```